

combined_cv_rover

scalars_rover_cv

key	value
skipped_empty_gt	0
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
step2_stats.n_models	9
step2_stats.spearman	1
step2_stats.kendall_b	1
step2_stats.spearman_exact_p_onetailed	2.7557e-06
step2_stats.kendall_exact_p_onetailed	2.7557e-06
step2_stats.p_method	exact
step2_stats.n_permutations	362880
step2_stats.min_attainable_p	2.7557e-06
step2_stats.pearson	0.9958
step2_stats.pearson_p	1.5886e-08
step2_stats.mean_wer_compression	0.7526
step3_determinism.identical_outputs	10495
step3_determinism.n	10495
step3_determinism.deterministic	True
step3_determinism.main_kwargs.substitute	False
step3_determinism.main_kwargs.normalize_final_letters	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	10500
step4_resolution_true.min_resolved_gap_wer	0.6214
step4_resolution_true.max_unresolved_gap_wer	0
step4_resolution_true.n_resolved	36
step4_resolution_true.n_pairs	36
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	10495
step4_resolution_pseudo.min_resolved_gap_wer	0.7957
step4_resolution_pseudo.max_unresolved_gap_wer	0
step4_resolution_pseudo.n_resolved	36
step4_resolution_pseudo.n_pairs	36

step0_cross_wer_rover_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.8	19.4	11.85	24.88	25.77	13.86	30.72	10.23
whisper_v3	15.8	0	14.77	15	25.5	26.23	17.06	31.71	13.88
whisper_v2	19.4	14.77	0	18.63	28.03	28.83	20.5	33.99	17.57
fastconformer	11.85	15	18.63	0	22.77	24.06	10.3	28.8	7.05
xlsr	24.88	25.5	28.03	22.77	0	31.68	23.49	31.1	22.29
w2vbert	25.77	26.23	28.83	24.06	31.68	0	25.33	37.2	23.32
hubert	13.86	17.06	20.5	10.3	23.49	25.33	0	29.58	9.23
wav2vec_sb	30.72	31.71	33.99	28.8	31.1	37.2	29.58	0	28.81
cohere	10.23	13.88	17.57	7.05	22.29	23.32	9.23	28.81	0

step0_err_corr_rover_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	1	0.493	0.4572	0.4468	0.3933	0.3559	0.3527	0.3807	0.4864
whisper_v3	0.493	1	0.6827	0.4367	0.4573	0.439	0.3487	0.4105	0.4442
whisper_v2	0.4572	0.6827	1	0.3904	0.4453	0.4221	0.3313	0.4116	0.4045
fastconformer	0.4468	0.4367	0.3904	1	0.41	0.3553	0.4155	0.4007	0.5511
xlsr	0.3933	0.4573	0.4453	0.41	1	0.4547	0.4217	0.6335	0.3863
w2vbert	0.3559	0.439	0.4221	0.3553	0.4547	1	0.3421	0.4296	0.3379
hubert	0.3527	0.3487	0.3313	0.4155	0.4217	0.3421	1	0.4044	0.3997
wav2vec_sb	0.3807	0.4105	0.4116	0.4007	0.6335	0.4296	0.4044	1	0.3594
cohere	0.4864	0.4442	0.4045	0.5511	0.3863	0.3379	0.3997	0.3594	1

step0_flags_rover_cv

pair	cross_wer	err_corr	note
seamless_m4t fastconformer	11.85	0.4468	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t hubert	13.86	0.3527	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t cohere	10.23	0.4864	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v3 cohere	13.88	0.4442	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer hubert	10.3	0.4155	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer cohere	7.05	0.5511	unusually similar hypotheses - suspect shared lab/data/teacher
hubert cohere	9.23	0.3997	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_rover_cv

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	10495	0.5832	0.5818	0.5573	0.5525	-0.0014	-0.0048
fastconformer	voter	10495	0.7057	0.7092	0.6727	0.674	0.0035	0.0013
hubert	voter	10495	0.7213	0.7056	0.6895	0.6709	-0.0158	-0.0186
seamless_m4t	voter	10495	0.8005	0.8062	0.7633	0.7675	0.0057	0.0042
whisper_v3	voter	10495	0.8694	0.8837	0.8273	0.8454	0.0143	0.018
whisper_v2	ranked_only(control)	10495	0.9054	0.9054	0.864	0.864	0	0
w2vbert	voter	10495	0.9212	0.9315	0.8768	0.8893	0.0104	0.0124
xlsr	ranked_only(control)	10495	0.9279	0.9279	0.8866	0.8866	0	0
wav2vec_sb	voter	10495	0.9518	0.9573	0.9137	0.9206	0.0054	0.0069

step2_ranking_rover_cv

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	10496	5.5591	1.5028	5.643	1.7637	1	1	-0.0839
fastconformer	voter	10496	7.1898	2.158	7.0657	2.2595	2	2	0.1241
hubert	voter	10495	7.8081	2.1291	9.0767	2.5375	3	3	-1.2687
seamless_m4t	voter	10495	10.7671	3.4897	9.8725	3.4809	4	4	0.8946
whisper_v3	voter	10495	14.7106	4.636	13.2142	4.3401	5	5	1.4964
whisper_v2	ranked_only	10495	18.4155	6.0072	16.5904	5.6871	6	6	1.8251
xlsr	ranked_only	10495	22.8174	6.433	21.4653	6.1227	7	7	1.3521
w2vbert	voter	10495	23.9822	6.5829	22.7165	6.3372	8	8	1.2657
wav2vec_sb	voter	10495	29.1187	9.6559	27.951	9.3329	9	9	1.1676

step3_model_dropout_rover_cv

model	times_filtered_out	times_in_pool	filtered_out_rate
wav2vec_sb	39018	73467	0.5311
w2vbert	29852	73467	0.4063
whisper_v3	13295	73467	0.181
seamless_m4t	8429	73467	0.1147
hubert	6275	73467	0.0854
fastconformer	4214	73466	0.0574
cohere	2518	73466	0.0343

step3_runs_rover_cv

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_rover_full_wtn
full_pool	7	10495	0.8566	0.0416	1

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_rover_full_wtn
LOO_minus_seamless_m4t	6	10495	0.8351	0.0348	1
LOO_minus_whisper_v3	6	10495	0.8309	0.0309	1
LOO_minus_fastconformer	6	10496	0.8407	0.0371	1
LOO_minus_w2vbert	6	10495	0.7747	0.0261	1
LOO_minus_hubert	6	10495	0.8411	0.0353	1
LOO_minus_wav2vec_sb	6	10495	0.7482	0.0253	1
LOO_minus_cohere	6	10496	0.8413	0.0409	1

step4_pairs_pseudo_rover_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
hubert < seamless_m4t	True	0.7957	1	True
xlsr < w2vbert	True	1.2512	1	True
cohere < fastconformer	True	1.4227	1	True
fastconformer < hubert	True	2.0128	1	True
fastconformer < seamless_m4t	False	2.8085	1	True
seamless_m4t < whisper_v3	True	3.3417	1	True
whisper_v3 < whisper_v2	True	3.3762	1	True
cohere < hubert	False	3.4355	1	True
hubert < whisper_v3	False	4.1374	1	True
cohere < seamless_m4t	False	4.2312	1	True
whisper_v2 < xlsr	True	4.8749	1	True
w2vbert < wav2vec_sb	True	5.2345	1	True
whisper_v2 < w2vbert	False	6.1261	1	True
fastconformer < whisper_v3	False	6.1502	1	True
xlsr < wav2vec_sb	False	6.4858	1	True
seamless_m4t < whisper_v2	False	6.7179	1	True
hubert < whisper_v2	False	7.5137	1	True
cohere < whisper_v3	False	7.5729	1	True
whisper_v3 < xlsr	False	8.2511	1	True
whisper_v3 < w2vbert	False	9.5024	1	True
fastconformer < whisper_v2	False	9.5264	1	True
cohere < whisper_v2	False	10.9492	1	True
whisper_v2 < wav2vec_sb	False	11.3606	1	True
seamless_m4t < xlsr	False	11.5928	1	True
hubert < xlsr	False	12.3885	1	True
seamless_m4t < w2vbert	False	12.8441	1	True
hubert < w2vbert	False	13.6398	1	True
fastconformer < xlsr	False	14.4013	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < wav2vec_sb	False	14.7369	1	True
fastconformer < w2vbert	False	15.6526	1	True
cohere < xlsr	False	15.8241	1	True
cohere < w2vbert	False	17.0753	1	True
seamless_m4t < wav2vec_sb	False	18.0786	1	True
hubert < wav2vec_sb	False	18.8743	1	True
fastconformer < wav2vec_sb	False	20.8871	1	True
cohere < wav2vec_sb	False	22.3098	1	True

step4_pairs_true_rover_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
fastconformer < hubert	True	0.6214	1	True
xlsr < w2vbert	True	1.1621	1	True
cohere < fastconformer	True	1.6296	1	True
cohere < hubert	False	2.2511	1	True
hubert < seamless_m4t	True	2.9664	1	True
fastconformer < seamless_m4t	False	3.5878	1	True
whisper_v3 < whisper_v2	True	3.7042	1	True
seamless_m4t < whisper_v3	True	3.9295	1	True
whisper_v2 < xlsr	True	4.3988	1	True
w2vbert < wav2vec_sb	True	5.1348	1	True
cohere < seamless_m4t	False	5.2174	1	True
whisper_v2 < w2vbert	False	5.561	1	True
xlsr < wav2vec_sb	False	6.2969	1	True
hubert < whisper_v3	False	6.8958	1	True
fastconformer < whisper_v3	False	7.5173	1	True
seamless_m4t < whisper_v2	False	7.6337	1	True
whisper_v3 < xlsr	False	8.103	1	True
cohere < whisper_v3	False	9.1469	1	True
whisper_v3 < w2vbert	False	9.2652	1	True
hubert < whisper_v2	False	10.6	1	True
whisper_v2 < wav2vec_sb	False	10.6958	1	True
fastconformer < whisper_v2	False	11.2215	1	True
seamless_m4t < xlsr	False	12.0325	1	True
cohere < whisper_v2	False	12.8511	1	True
seamless_m4t < w2vbert	False	13.1947	1	True
whisper_v3 < wav2vec_sb	False	14.4	1	True
hubert < xlsr	False	14.9989	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
fastconformer < xlsr	False	15.6203	1	True
hubert < w2vbert	False	16.161	1	True
fastconformer < w2vbert	False	16.7824	1	True
cohere < xlsr	False	17.2499	1	True
seamless_m4t < wav2vec_sb	False	18.3295	1	True
cohere < w2vbert	False	18.4121	1	True
hubert < wav2vec_sb	False	21.2958	1	True
fastconformer < wav2vec_sb	False	21.9172	1	True
cohere < wav2vec_sb	False	23.5469	1	True

step5_baselines_rover_cv

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.0531
full ensemble (biased)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.7173
single model: seamless_m4t	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.8048
LOO ensemble matched to n-1 (vs seamless_m4t)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.0729
single model: whisper_v3	8	0.8333	0.7143	0.0077	0.0071	exact	40320	2.4802e-05	5.1675
LOO ensemble matched to n-1 (vs whisper_v3)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.9977
single model: fastconformer	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7331
LOO ensemble matched to n-1 (vs fastconformer)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1693
single model: w2vbert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	13.2048
LOO ensemble matched to n-1 (vs w2vbert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.0266
single model: hubert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.003
LOO ensemble matched to n-1 (vs hubert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.0262
single model: wav2vec_sb	8	0.9286	0.8571	0.0011	0.0009	exact	40320	2.4802e-05	17.8119
LOO ensemble matched to n-1 (vs wav2vec_sb)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.0388
single model: cohere	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7447
LOO ensemble matched to n-1 (vs cohere)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1743

step6_contribution_rover_cv

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	0.2762	0	True	True	0	0
whisper_v3	-0.1935	0	False	True	0	0
fastconformer	0.5335	0	True	True	0	0
w2vbert	-0.1315	0	False	True	0	0
hubert	0.6557	0	True	True	0	0

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
wav2vec_sb	-0.1785	0	False	True	0	0
cohere	0.866	0	True	True	0	0

step6_pools_rover_cv

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+fastconformer+hubert+cohere	4	4.5334	1.2955	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+cohere	5	4.6029	1.2993	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	4.6818	1.3442	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+cohere	5	4.6893	1.3031	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	4.7852	1.4561	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+cohere	6	4.8002	1.4066	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	6	4.8472	1.4751	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+cohere	4	4.9204	1.4363	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	7	4.9787	1.5599	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+cohere	4	5.0069	1.5158	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.8931	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	5.0444	1.4538	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	5.0876	1.5112	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	5.1759	1.5603	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+wav2vec_sb+cohere	4	5.1947	1.7095	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.8931	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	5.2549	1.6482	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+w2vbert+hubert+cohere	4	5.2942	1.5938	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.5669	cohere fastconformer
fastconformer+hubert+cohere	3	5.377	2.2531								
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	5.39	1.5439	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+hubert+cohere	4	5.4033	1.625	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6296	cohere fastconformer
seamless_m4t+hubert+wav2vec_sb+cohere	4	5.4501	1.8064	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.5669	cohere fastconformer
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	5.4933	1.6116	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+w2vbert+cohere	4	5.4953	1.6223	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5555	fastconformer hubert
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	5.499	1.6326	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	5.5085	1.7193	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5555	fastconformer hubert
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	5.5121	1.658	1	2.7557e-06	1	2.7557e-06	0	0	0	

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	5.5855	1.6413	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	6	5.6344	1.679	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+cohere	5	5.6607	1.5839	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert	4	5.6775	1.6828	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+cohere	4	5.6775	1.6573	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb+cohere	5	5.6813	1.6379	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert	4	5.7338	1.7418	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	5.7338	1.6946	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	5.7789	1.6706	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	5.7958	1.7113	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	5.8447	1.8019	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	5.8898	1.9236	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+cohere	3	5.9365	2.4082								
seamless_m4t+hubert+cohere	3	5.9778	2.7667								
whisper_v3+w2vbert+hubert+cohere	4	5.9856	1.8087	0.9167	0.0007	0.8333	0.0004	3	3	5.1458	cohere seamless_m4t
whisper_v3+hubert+wav2vec_sb+cohere	4	6.1265	2.0309	0.9167	0.0007	0.8333	0.0004	3	3	5.1458	cohere seamless_m4t
whisper_v3+fastconformer+wav2vec_sb+cohere	4	6.1491	1.9518	0.9167	0.0007	0.8333	0.0004	3	3	3.5145	fastconformer seamless_m4t
whisper_v3+fastconformer+w2vbert+cohere	4	6.1942	1.8575	0.9167	0.0007	0.8333	0.0004	3	3	3.5145	fastconformer seamless_m4t
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	6.2091	1.9069	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.9571	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	6.3239	1.9096	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert	3	6.3739	2.8823								
fastconformer+w2vbert+wav2vec_sb+cohere	4	6.4441	2.0933	0.8667	0.0023	0.7778	0.0012	4	4	5.208	cohere seamless_m4t
whisper_v3+fastconformer+cohere	3	6.4997	2.5607								
whisper_v3+fastconformer+w2vbert+hubert	4	6.5473	2.0416	0.95	0.0002	0.8889	0.0001	2	2	3.5162	fastconformer seamless_m4t
whisper_v3+hubert+cohere	3	6.5523	2.9234								
whisper_v3+fastconformer+hubert+wav2vec_sb	4	6.6055	2.1568	0.95	0.0002	0.8889	0.0001	2	2	3.5162	fastconformer seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	6.6243	2.0267	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
w2vbert+hubert+wav2vec_sb+cohere	4	6.7408	2.2828	0.9167	0.0007	0.8333	0.0004	3	3	5.2097	cohere seamless_m4t
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	6.8479	1.9932	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+whisper_v3+w2vbert+cohere	4	6.9728	2.071	0.95	0.0002	0.8889	0.0001	2	2	2.2511	cohere hubert

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	6.9803	2.1808	0.95	0.0002	0.8889	0.0001	2	2	2.2511	cohere hubert
whisper_v3+fastconformer+hubert	3	7.0085	3.0268								
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	7.0903	2.2086	0.9333	0.0004	0.8333	0.0004	3	3	3.8718	seamless_m4t whisper_v3
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	7.2556	2.1854	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+wav2vec_sb	4	7.2612	2.4825	0.95	0.0002	0.8889	0.0001	2	2	3.5789	fastconformer seamless_m4t
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	7.3044	2.1812	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.6219	fastconformer hubert
seamless_m4t+whisper_v3+w2vbert+hubert	4	7.3683	2.2413	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+cohere	3	7.3821	3.2195								
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	7.4209	2.4433	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	3.6491	whisper_v3 whisper_v2
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	7.4247	2.349	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	7.4472	2.2268	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	7.6914	2.4661	0.9167	0.0007	0.8333	0.0004	3	3	7.4596	fastconformer whisper_v3
seamless_m4t+whisper_v3+cohere	3	7.7294	3.0926								
w2vbert+hubert+cohere	3	7.752	3.6275								
fastconformer+wav2vec_sb+cohere	3	7.812	3.6028								
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	7.915	2.6393	0.95	0.0002	0.8889	0.0001	2	2	6.8382	hubert whisper_v3
seamless_m4t+whisper_v3+fastconformer	3	8.1181	3.3005								
fastconformer+w2vbert+hubert	3	8.21	3.6947								
hubert+wav2vec_sb+cohere	3	8.2645	3.8685								
seamless_m4t+whisper_v3+hubert	3	8.272	3.7328								
seamless_m4t+w2vbert+cohere	3	8.4973	3.6594								
whisper_v3+w2vbert+wav2vec_sb+cohere	4	8.6515	2.7298	0.8833	0.0015	0.7778	0.0012	4	4	5.1458	cohere seamless_m4t
fastconformer+hubert+wav2vec_sb	3	8.7623	3.964								
seamless_m4t+wav2vec_sb+cohere	3	8.8352	4.0438								
seamless_m4t+fastconformer+w2vbert	3	9.0718	3.9043								
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	9.2339	2.96	0.9333	0.0004	0.8333	0.0004	3	3	3.7067	whisper_v3 whisper_v2
seamless_m4t+w2vbert+hubert	3	9.4191	4.3567								
whisper_v3+w2vbert+hubert+wav2vec_sb	4	9.4593	3.138	0.9667	8.2672e-05	0.8889	0.0001	2	2	3.7067	whisper_v3 whisper_v2
seamless_m4t+fastconformer+wav2vec_sb	3	9.4736	4.3016								
seamless_m4t+hubert+wav2vec_sb	3	9.9335	4.6864								
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	10.0004	3.1677	0.95	0.0002	0.8889	0.0001	2	2	7.5785	seamless_m4t whisper_v2
whisper_v3+wav2vec_sb+cohere	3	10.3128	4.5008								
whisper_v3+w2vbert+cohere	3	10.3372	4.1564								
whisper_v3+fastconformer+w2vbert	3	10.8535	4.3666								
whisper_v3+fastconformer+wav2vec_sb	3	10.9323	4.6898								
whisper_v3+w2vbert+hubert	3	11.1163	4.7974								
whisper_v3+hubert+wav2vec_sb	3	11.3304	5.0228								
seamless_m4t+whisper_v3+w2vbert	3	11.9668	4.7194								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+whisper_v3+wav2vec_sb	3	12.0982	5.1517								
w2vbert+wav2vec_sb+cohere	3	12.9412	5.4715								
fastconformer+w2vbert+wav2vec_sb	3	13.2543	5.4714								
w2vbert+hubert+wav2vec_sb	3	13.7071	5.9405								
seamless_m4t+w2vbert+wav2vec_sb	3	14.6121	6.06								
whisper_v3+w2vbert+wav2vec_sb	3	16.2643	6.4665								

step6_wrong_pairs_rover_cv

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	seamless_m4t whisper_v2	seamless_m4t	whisper_v2	10.8403	18.4189	17.6212	16.588	7.5785	1.0333	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	7.2526	14.7122	14.5305	13.964	7.4596	0.5666	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	hubert whisper_v3	hubert	whisper_v3	7.874	14.7122	15.8862	14.6747	6.8382	1.2115	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7688	13.4216	11.4158	5.2097	2.0058	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7671	12.4382	10.8057	5.208	1.6325	True
whisper_v3+w2vbert+hubert+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.6229	10.7688	10.828	10.3022	5.1458	0.5258	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.6229	10.7688	14.9597	11.5663	5.1458	3.3934	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.6229	10.7688	11.0391	10.5364	5.1458	0.5026	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	16.1426	13.964	3.9434	2.1787	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	17.0255	14.6747	3.9434	2.3508	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.8403	14.7122	15.7974	13.6545	3.8718	2.1429	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	18.1463	16.8591	3.7067	1.2872	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	19.104	17.4267	3.7067	1.6773	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7698	18.4189	18.3013	16.588	3.6491	1.7133	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7698	18.4189	16.5661	16.5621	3.6491	0.0039	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7698	18.4189	17.9696	16.5845	3.6491	1.3851	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7688	14.2196	11.9741	3.5789	2.2455	True
whisper_v3+fastconformer+w2vbert+hubert	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7688	11.9868	10.8189	3.5162	1.1679	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7688	15.8269	12.1348	3.5162	3.6921	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7688	12.0745	10.9884	3.5162	1.0861	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7671	10.9427	10.2307	3.5145	0.712	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7671	10.8408	10.1773	3.5145	0.6636	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.2526	10.7671	13.1715	10.8057	3.5145	2.3658	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	14.6244	11.9741	2.9571	2.6502	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.623	10.7879	2.9571	0.8351	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.359	10.3939	2.9571	0.9651	True
whisper_v3+fastconformer+w2vbert+hubert	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.9375	10.8189	2.8948	2.1186	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.8953	10.9884	2.8948	1.9069	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	17.2223	12.7364	2.8948	4.4859	True
whisper_v3+w2vbert+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.7769	10.3022	2.8948	2.4747	True
whisper_v3+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.6363	10.5364	2.8948	2.0999	True
w2vbert+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	14.5255	11.4158	2.8948	3.1096	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
fastconformer+w2vbert+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7671	10.3488	10.3104	2.8931	0.0384	True
fastconformer+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7671	10.5806	10.4749	2.8931	0.1057	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere hubert	cohere	hubert	5.6229	7.874	10.9142	10.2586	2.2511	0.6556	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.6229	7.874	11.0194	10.1995	2.2511	0.8199	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8081	12.4382	9.5941	2.249	2.8441	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.6229	7.8117	13.4693	10.1802	2.1887	3.2891	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.6229	7.8117	14.9597	11.359	2.1887	3.6008	True
whisper_v3+fastconformer+w2vbert+cohere	4	cohere hubert	cohere	hubert	5.6229	7.8081	9.9875	9.5105	2.1851	0.477	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.6229	7.8081	10.0799	9.3977	2.1851	0.6822	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	13.4216	8.2689	1.6307	5.1527	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.1114	7.9519	1.6307	1.1595	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.2133	7.0643	1.6307	0.149	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.8933	7.1315	1.6307	0.7618	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.2149	7.5483	1.6307	1.6666	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	10.9142	8.169	1.6296	2.7451	True
seamless_m4t+whisper_v3+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	8.1441	7.1692	1.6296	0.9749	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	11.0194	8.0904	1.6296	2.929	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	9.3513	7.1239	1.5669	2.2274	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	13.4693	8.2851	1.5669	5.1842	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	9.8824	7.212	1.5669	2.6704	True
whisper_v3+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	10.828	7.5348	1.5669	3.2932	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	14.9597	9.3618	1.5669	5.5979	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.1898	11.0391	7.6012	1.5669	3.4378	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	10.5205	10.243	0.6219	0.2775	True
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	12.3454	10.4545	0.5591	1.8909	True
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	12.3632	10.3202	0.5591	2.0429	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	14.5305	10.4949	0.5591	4.0356	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	15.8269	11.623	0.5591	4.2039	True
seamless_m4t+fastconformer+w2vbert+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	9.619	9.0357	0.5555	0.5834	True
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	9.7756	8.9888	0.5555	0.7868	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	10.9427	9.5105	0.5555	1.4322	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	10.8408	9.3977	0.5555	1.4431	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	13.1715	9.5941	0.5555	3.5775	True

step7_floor_rover_cv

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.8011	0.5528	0.3377	2088	0.5703
hubert	0.7598	0.6709	0.5889	2521	0.6972
fastconformer	0.757	0.6741	0.4772	2551	0.6872
seamless_m4t	0.6598	0.7675	0.5965	3570	0.7831
whisper_v3	0.5688	0.8454	0.7239	4525	0.8552

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
whisper_v2	0.4936	0.864	0.7592	5315	0.8743
xlsr	0.4227	0.8866	0.8038	6059	0.9067
w2vbert	0.3718	0.8893	0.8221	6593	0.9075
wav2vec_sb	0.317	0.9206	0.8732	7168	0.9372